

04_axis_bias_compression

```
source("R/00_setup.R")
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
model_coords <- readRDS("model_coords.rds")
coords       <- readRDS("country_wave_coords.rds")

latest_gt <- coords %>%
  transmute(country = as.character(iso), wave = as.numeric(wave),
            gt1 = Dim1, gt2 = Dim2) %>%
  group_by(country) %>% filter(wave == max(wave)) %>% ungroup() %>%
  select(country, gt1, gt2)

#' Build the latest-wave frame for a condition. C2 carries a wave column and is
#' reduced to each country's most recent; C1 has none and is compared directly to
#' the most recent truth.

get_latest <- function(cond) {
  df <- model_coords %>% filter(condition == cond) %>% mutate(country = as.character(country))
  if ("wave" %in% names(df) && any(!is.na(df$wave))) {
    df <- df %>% mutate(wave = as.numeric(wave)) %>%
```

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    group_by(model, country) %>% filter(wave == max(wave, na.rm = TRUE)) %>% ungroup()
  }
  df %>% select(model, country, Dim1, Dim2) %>%
    inner_join(latest_gt, by = "country") %>%
    mutate(d1 = Dim1 - gt1, d2 = Dim2 - gt2)
}

#' Spread ratio: cross-country SD of model coordinates over cross-country SD of
#' the truth, on one axis. Below one is compression, above one expansion.

boot_spread <- function(df, axis) {
  mcol <- if (axis == 1) "Dim1" else "Dim2"
  gcol <- if (axis == 1) "gt1" else "gt2"
  Wm <- df %>% select(model, country, val = all_of(mcol)) %>%
    pivot_wider(names_from = model, values_from = val) %>% arrange(country)
  G <- df %>% distinct(country, .keep_all = TRUE) %>%
    select(country, gtv = all_of(gcol)) %>% arrange(country)
  stopifnot(identical(Wm$country, G$country))
  M <- as.matrix(Wm[, -1]); g <- G$gtv; n <- nrow(M)
  set.seed(SEED)
  out <- matrix(NA_real_, B, ncol(M), dimnames = list(NULL, colnames(M)))
  for (b in seq_len(B)) {
    idx <- sample.int(n, n, replace = TRUE)
    out[b, ] <- apply(M[idx, , drop = FALSE], 2, sd, na.rm = TRUE) / sd(g[idx], na.rm = TRUE)
  }
  out
}

#' ## Run both conditions
#'
#' Dimension 1 is traditional vs. secular-rational, where a negative bias means
#' the model places countries too far toward the traditional pole. Dimension 2 is
#' survival vs. self-expression, where a positive bias means too far toward
#' self-expression.

res <- list()
for (cond in c("C1", "C2")) {
  df <- get_latest(cond)
  cat(sprintf("\n\n##### CONDITION %s (n countries = %d) #####\n",
    cond, n_distinct(df$country)))

  b1 <- boot_paired(df, "d1")

```

```

b2 <- boot_paired(df, "d2")
report_ci(b1, sprintf("%s Dim1 signed bias (- = too traditional)", cond))
report_pairwise(b1, sprintf("%s Dim1 bias", cond))
report_ci(b2, sprintf("%s Dim2 signed bias (+ = too self-expressive)", cond))
report_pairwise(b2, sprintf("%s Dim2 bias", cond))

s1 <- boot_spread(df, 1)
s2 <- boot_spread(df, 2)
report_ci(s1, sprintf("%s Dim1 spread ratio (model SD / truth SD)", cond), vs = 1)
report_ci(s2, sprintf("%s Dim2 spread ratio (model SD / truth SD)", cond), vs = 1)

cat(sprintf("\n=== %s raw SDs, model vs truth ===\n", cond))
df %>% group_by(model) %>%
  summarise(sd_model_D1 = round(sd(Dim1), 3), sd_truth_D1 = round(sd(gt1), 3),
            sd_model_D2 = round(sd(Dim2), 3), sd_truth_D2 = round(sd(gt2), 3),
            .groups = "drop") %>% print()

res[[cond]] <- list(bias_d1 = b1, bias_d2 = b2, spread_d1 = s1, spread_d2 = s2)
}

```

CONDITION C1 (n countries = 40)

=== C1 Dim1 signed bias (- = too traditional) ===

	X2.5.	X50.	X97.5.	vs_ref
Qwen/Qwen3.6-Plus	0.018	0.067	0.114	ABOVE
claude-opus-4-8	-0.098	-0.049	-0.001	BELOW
gemini-3.6-flash	-0.093	-0.047	-0.004	BELOW
gpt-5.5	-0.092	-0.046	-0.002	BELOW

=== C1 Dim1 bias: pairwise difference (row - col), 95% CI; * = excludes 0 ===

		lo2.5	median	hi97.5	sig
Qwen/Qwen3.6-Plus	vs claude-opus-4-8	0.083	0.116	0.146	*
Qwen/Qwen3.6-Plus	vs gemini-3.6-flash	0.078	0.113	0.147	*
Qwen/Qwen3.6-Plus	vs gpt-5.5	0.089	0.113	0.137	*
claude-opus-4-8	vs gemini-3.6-flash	-0.053	-0.002	0.047	
claude-opus-4-8	vs gpt-5.5	-0.044	-0.003	0.039	
gemini-3.6-flash	vs gpt-5.5	-0.018	-0.001	0.019	

=== C1 Dim2 signed bias (+ = too self-expressive) ===

	X2.5.	X50.	X97.5.	vs_ref
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Qwen/Qwen3.6-Plus	0.067	0.118	0.169	ABOVE
claude-opus-4-8	0.010	0.043	0.076	ABOVE
gemini-3.6-flash	-0.029	0.000	0.031	n.s.
gpt-5.5	0.048	0.085	0.124	ABOVE

=== C1 Dim2 bias: pairwise difference (row - col), 95% CI; * = excludes 0 ===

		lo2.5	median	hi97.5	sig
Qwen/Qwen3.6-Plus	vs claude-opus-4-8	0.041	0.075	0.111	*
Qwen/Qwen3.6-Plus	vs gemini-3.6-flash	0.081	0.118	0.155	*
Qwen/Qwen3.6-Plus	vs gpt-5.5	0.005	0.033	0.061	*
claude-opus-4-8	vs gemini-3.6-flash	0.025	0.043	0.060	*
claude-opus-4-8	vs gpt-5.5	-0.063	-0.042	-0.021	*
gemini-3.6-flash	vs gpt-5.5	-0.105	-0.085	-0.065	*

=== C1 Dim1 spread ratio (model SD / truth SD) ===

	X2.5.	X50.	X97.5.	vs_ref
Qwen/Qwen3.6-Plus	0.779	0.887	1.006	n.s.
claude-opus-4-8	0.585	0.653	0.731	BELOW
gemini-3.6-flash	0.979	1.094	1.233	n.s.
gpt-5.5	0.901	1.024	1.168	n.s.

=== C1 Dim2 spread ratio (model SD / truth SD) ===

	X2.5.	X50.	X97.5.	vs_ref
Qwen/Qwen3.6-Plus	0.666	0.774	0.927	BELOW
claude-opus-4-8	0.801	0.889	0.999	BELOW
gemini-3.6-flash	0.881	0.966	1.064	n.s.
gpt-5.5	0.811	0.909	1.039	n.s.

=== C1 raw SDs, model vs truth ===

A tibble: 4 x 5

model	sd_model_D1	sd_truth_D1	sd_model_D2	sd_truth_D2
<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1 Qwen/Qwen3.6-Plus	0.289	0.327	0.262	0.338
2 claude-opus-4-8	0.213	0.327	0.301	0.338
3 gemini-3.6-flash	0.358	0.327	0.327	0.338
4 gpt-5.5	0.335	0.327	0.307	0.338

CONDITION C2 (n countries = 40)

=== C2 Dim1 signed bias (- = too traditional) ===

	X2.5.	X50.	X97.5.	vs_ref
Qwen/Qwen3.6-Plus	0.006	0.059	0.107	ABOVE

claude-opus-4-8	-0.094	-0.039	0.012	n.s.
gemini-3.6-flash	-0.112	-0.067	-0.027	BELOW
gpt-5.5	-0.091	-0.047	-0.005	BELOW

=== C2 Dim1 bias: pairwise difference (row - col), 95% CI; * = excludes 0 ===

		lo2.5	median	hi97.5	sig
Qwen/Qwen3.6-Plus	vs claude-opus-4-8	0.063	0.098	0.132	*
Qwen/Qwen3.6-Plus	vs gemini-3.6-flash	0.086	0.126	0.162	*
Qwen/Qwen3.6-Plus	vs gpt-5.5	0.080	0.106	0.130	*
claude-opus-4-8	vs gemini-3.6-flash	-0.025	0.028	0.078	
claude-opus-4-8	vs gpt-5.5	-0.034	0.007	0.049	
gemini-3.6-flash	vs gpt-5.5	-0.038	-0.020	-0.001	*

=== C2 Dim2 signed bias (+ = too self-expressive) ===

	X2.5.	X50.	X97.5.	vs_ref
Qwen/Qwen3.6-Plus	0.036	0.088	0.136	ABOVE
claude-opus-4-8	0.020	0.052	0.085	ABOVE
gemini-3.6-flash	-0.043	-0.013	0.017	n.s.
gpt-5.5	0.036	0.073	0.110	ABOVE

=== C2 Dim2 bias: pairwise difference (row - col), 95% CI; * = excludes 0 ===

		lo2.5	median	hi97.5	sig
Qwen/Qwen3.6-Plus	vs claude-opus-4-8	-0.001	0.036	0.073	
Qwen/Qwen3.6-Plus	vs gemini-3.6-flash	0.060	0.101	0.141	*
Qwen/Qwen3.6-Plus	vs gpt-5.5	-0.020	0.015	0.049	
claude-opus-4-8	vs gemini-3.6-flash	0.048	0.065	0.081	*
claude-opus-4-8	vs gpt-5.5	-0.040	-0.021	-0.001	*
gemini-3.6-flash	vs gpt-5.5	-0.106	-0.086	-0.065	*

=== C2 Dim1 spread ratio (model SD / truth SD) ===

	X2.5.	X50.	X97.5.	vs_ref
Qwen/Qwen3.6-Plus	0.774	0.881	1.001	n.s.
claude-opus-4-8	0.564	0.652	0.752	BELOW
gemini-3.6-flash	1.006	1.106	1.225	ABOVE
gpt-5.5	0.885	1.003	1.145	n.s.

=== C2 Dim2 spread ratio (model SD / truth SD) ===

	X2.5.	X50.	X97.5.	vs_ref
Qwen/Qwen3.6-Plus	0.654	0.760	0.907	BELOW
claude-opus-4-8	0.856	0.946	1.043	n.s.
gemini-3.6-flash	0.910	0.994	1.095	n.s.
gpt-5.5	0.892	0.981	1.105	n.s.

```
=== C2 raw SDs, model vs truth ===
```

```
# A tibble: 4 x 5
```

	model	sd_model_D1	sd_truth_D1	sd_model_D2	sd_truth_D2
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	Qwen/Qwen3.6-Plus	0.287	0.327	0.257	0.338
2	claude-opus-4-8	0.213	0.327	0.319	0.338
3	gemini-3.6-flash	0.361	0.327	0.337	0.338
4	gpt-5.5	0.328	0.327	0.331	0.338

```
saveRDS(res, "axis_bias_spread.rds")
```